Diffusion Inference Runtime

Deep Dive into SGLang — Chapter 31

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Where this chapter sits

This is the chapter that turns "call the model in a loop" into a request-owned continuation record the runtime must keep aligned.

- Volume I served *token streams*: a request appends one decoded token to a sequence, prefill then decode.
- A diffusion request instead *rewrites a latent* many times under a scheduler timestep, then decodes it to an image, video, mesh, or audio artifact.
- One trace carries the chapter: a `Qwen-Image` text-to-image request — prompt "a red mug on a wooden desk," seed 1337, 1024×1024, 30 denoising steps.

The unit of work changed: not a token suffix, but a latent, timestep, conditioning, generator, and component version that must name one request.

Roadmap

From token streams to denoising state

The staged pipeline

Loop, decode, and the invariant

Tradeoffs and evidence

From token streams to denoising state

The continuation record is a tuple, not a token

Definition (Diffusion pipeline state)
For request r at timestep t,

$$X_t(r) = (C_t, z_t, c, \sigma_t, g_{r,t}, o, r_{\text{params}}),$$

loaded components, latent, conditioning, scheduler state, generator state, output-shape metadata, and request parameters.

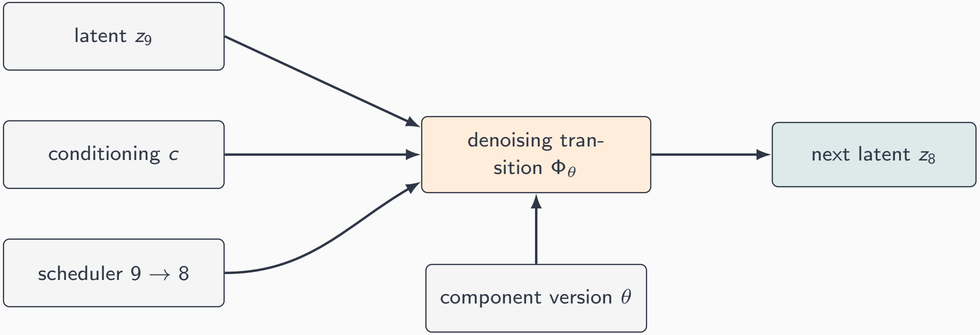
The repeated transition is one *denoising step*:

$$z_{t-1} = \Phi_\theta(z_t, c, t).$$

- Φ_θ is denoiser prediction *plus* the scheduler update rule that turns it into z_{t-1}.
- No sampled token, no KV-cache append — the carried state is a latent plus a scheduler position.

The chapter's one claim: a shape-correct latent update can still be wrong if it uses the wrong timestep, prompt, VAE scale, or stale component.

A denoising step is a state-aligned transition



The latent, timestep, conditioning, and active component version must all name the same request trajectory before the next latent is legal.

The staged pipeline

The request record carries identity across stages

The runtime object is the request batch, `Req`. Stages read the fields they own and hand the batch on — identity never detaches.

Book state	Representative `Req` owner
Request parameters	`sampling_params`
Conditioning	`prompt_embeds`, masks
Latent state	`latents`, `raw_latent_shape`
Scheduler trajectory	`timesteps`, `sigmas`, `step_index`
Reproducibility	`seed`, `generator`
Component / output	`modules`, `output`

A correct run changes many fields, but never which request owns the prompt, latent, scheduler, generator, component version, and output record.

Admission chooses exactly one trajectory interface

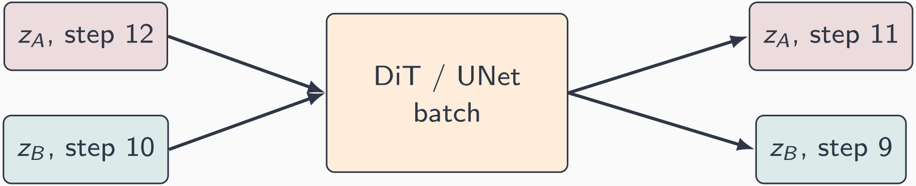
Before batching, the timestep-preparation stage binds the request to one scheduler trajectory.

- Custom `timesteps` *and* custom `sigmas` together are rejected — two competing ways to define the trajectory.
- The stage inspects `set_timesteps`' signature before passing `timesteps`, `sigmas`, or `n_tokens`.
- Otherwise it falls back to the plain step-count path.

This front-loads a decision token serving makes incrementally: every admitted row already has one schedule the scheduler can actually construct.

Loop, decode, and the invariant

The denoising loop batches by readiness, not by identity



- Rows share latent shape and the denoiser call, but sit at *different* scheduler positions.
- Each row must still carry its own conditioning, sigmas, seed, and `step_index`.

Batch membership cannot replace per-request timestep state.

Decode is part of the trajectory, not a postprocess

A VAE decode preserves the latent-to-media *meaning* of the request.

- Latent scale, shift, channel layout, tiling/slicing, and dtype can all change the output.
- Decode with another pipeline's scale or shift and the tensor stays the right image shape while no longer matching the denoiser trajectory.
- Image edit, image-to-video, audio, and mesh decode boundaries differ, but the ownership rule is unchanged.

The final artifact must be assembled from the same request trajectory that produced the final latent.

The step-alignment invariant

Invariant (Diffusion step-alignment)
For every request and every denoising step, the scheduler timestep, latent, conditioning, generator state, and component versions must refer to one logical request and advance as one transition:

$$S_{r,k} = (\tau_k, z_{r,k}, c_r, g_{r,k}, v_{r,k}).$$

- Swap only the latent pointers of two shape-matched rows: the VAE still decodes a plausible image, but the seed and trajectory no longer match.
- The failure is *not* an exception — it is silent state divergence a latent-norm or checksum ledger would catch.

Shape compatibility is necessary; request identity across every field is what makes the transition semantically valid.

Tradeoffs and evidence

Latency is loop-shaped, not stream-shaped

$$T_{\text{request}} = T_{\text{condition}} + \sum_{t \in \mathcal{T}} T_{\text{denoise}}(t) + T_{\text{decode}} + T_{\text{post}}.$$

- Token serving streams after the first token, so TTFT is special. Diffusion usually completes the whole trajectory before an artifact is useful.
- Pressure shifts from inter-token time to *step latency, component residency, and decode cost*.
- The same equation is a validity check: a residency win is not comparable if the next decode observes a different VAE scale or dtype.

Comparing two runs is meaningful only when every term names the same work under a fixed protocol.

What to carry forward

1. Diffusion serving is a **request-owned continuation record**, not "a model called in a loop."
2. $z_{t-1} = \Phi_\theta(z_t, c, t)$ is a map from notation to review: denoiser prediction plus scheduler update.
3. **Admission** binds one trajectory interface; **stages** carry identity; **decode** owns the latent-to-media convention.
4. The **step-alignment invariant** rejects shape-correct but identity-broken transitions — silent divergence, not exceptions.
5. **Latency is loop-shaped**: compare only under a fixed protocol where every term names the same work.

Next: from denoising state to acceleration — how residency, caching, and transfer policy cut cost while preserving the same continuation record.